

Niharikaa Agarwal

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Personal Profile

Computer Science undergraduate with strong Data Structures Algorithms foundation (300+ LeetCode problems solved). Experienced in backend development using Django and building AI-driven and game-based systems with applied algorithms (Minimax, Dijkstra).

Education

Graphic Era Deemed to be University

Dehradun, Uttarakhand, India

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2023 – Present

- Current CGPA: **8.12 (on a scale of 10)**
- Active member of the National Service Scheme (NSS)

Ann Mary School

Dehradun, Uttarakhand, India

Secondary and Higher Secondary Education – ICSE/ISC Board

2023

- Completed Class X (ICSE) and Class XII (ISC) with a focus on science and computer applications
- Studied core subjects: Physics, Chemistry, Mathematics, and Computer Science

Projects

The Great Heist – 2D Grid-Based Strategy Game

Unity, C#, Tilemaps, Algorithms (Minimax, Dijkstra), Photon PUN2

Jun 2025 – Oct 2025

- Architected and developed a 2D grid-based stealth strategy game in Unity using C#, supporting AI and multiplayer gameplay modes.
- Implemented adversarial AI using Minimax algorithm with heuristic evaluation and Dijkstra's algorithm for optimal pathfinding on dynamic grids.
- Designed a modular Tilemap-based grid system for walkable nodes, obstacles, collectibles, power-ups, and goal zones with state transitions.
- Engineered gameplay logic for coin collection, power-ups, scoring system, and win-condition validation using event-driven scripting.
- Currently integrating real-time multiplayer using Photon Unity Networking (PUN2).
- Deployed cross-platform builds: WebGL Version | Android Build.
- GitHub:** github.com/niharikaa0124/The-Great-Heist

Security-Based Hospital Management System

Python, Django, PostgreSQL, HTML, CSS, JavaScript

Apr 2025 – Jun 2025

- Designed and developed a full-stack hospital management system using Django following MVC architecture.
- Implemented secure role-based authentication and authorization for Admin, Doctor, and Patient users.
- Modeled relational database schemas in PostgreSQL for patients, appointments, medical history, and doctor records.
- Optimized backend using Django ORM for efficient CRUD operations and reduced redundancy.
- Built REST-like backend modules for appointments, user management, and record tracking.
- Developed responsive frontend using HTML, CSS, and JavaScript.
- GitHub:** github.com/niharikaa0124/hospital-management-system

Job Portal Web Application

Python, Django, FastAPI, React.js, SQLite, HTML, CSS, JavaScript, Django ORM, SMTP

Oct 2025 – Dec 2025

- Built a full-stack Job Portal using Django (MVT architecture) and React.js with role-based access for job seekers and recruiters.
- Implemented OTP-based authentication using Django email backend for secure login and verification.
- Developed RESTful APIs using FastAPI for job posting, application management, user profiles, and recruiter operations.
- Built responsive and interactive frontend interfaces using React.js, enabling seamless job search, filtering, and application workflows.
- Developed job posting, browsing, filtering, and bulk job posting features integrated with Django ORM and FastAPI services.
- Added resume upload functionality using Django FileField with proper media handling.
- Integrated frontend and backend components to deliver a scalable and maintainable full-stack application.
- GitHub:** github.com/niharikaa0124/django-job-portal-fullstack

Competitive Programming & DSA

- Solved 300+ LeetCode problems (mostly Medium level) across Arrays, Graphs, Trees, Dynamic Programming, and Backtracking.
- Proficient in core algorithms including BFS, DFS, Dijkstra's Algorithm, Greedy Techniques, Sliding Window, and Dynamic Programming.
- Regular coding contest participant with consistent focus on problem-solving and time-complexity optimization.

Technical Skills

Languages	Java, Python, C#
Web Technologies	Django, React.js, JavaScript
Databases	MySQL, PostgreSQL
Tools	Git, GitHub, Unity
Core Subjects	OOPS, DBMS, Operating Systems